Project Design Phase-II
Technology Stack (Architecture & Stack)

Date	31 January 2026
Team ID	[Add Team ID]
Project Name	Global Fuel Prices & Taxation Analysis Dashboard
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the technologies, platforms, databases, BI tools, and integrations that you will use to build your Business Intelligence solution. A well-chosen technology stack helps ensure that the solution is scalable, maintainable, secure, and suitable for data analysis and reporting requirements.

S.No	Component	Description	Technology	Justification / Purpose
1.	Programming / Query Language	Language used to query, filter, transform, aggregate, and analyze data.	Tableau calculated fields (CASE, STDEV, AVG); Python (pandas); HTML, CSS, JavaScript	Calculated fields create Subsidy Score, Avg Fuel Price and Diesel Volatility. Python was used to check and summarise data for the website, and HTML/CSS/JS build the project site.
2.	Database / Data Warehouse	Warehouse Used to store and manage structured business data for analysis and reporting.	CSV file (global_fuel_prices_Dataset.csv)	A single flat file holds all 27,468 records, so no database server is needed for this analysis.
3.	Data Ingestion	Used to load data from source files or external systems into the analytical environment.	Tableau Data Connection (Text file)	Loads the CSV directly into Tableau Desktop with automatic data types for date, country and

				numbers.
4.	Data Transformation and Business Logic	Tools/Logic used to clean, join, filter, aggregate, data for analysis and calculate KPIs, metrics, business rules, and analytical measures.	Tableau Desktop (calculated fields, filters, sorting)	Used to clean the view, create calculated fields and build the average, volatility and forecast measures.
5.	Data Visualization / Analytics	Used to create interactive dashboards, charts, reports, KPIs, and visual analysis.	Tableau Desktop and Tableau Public; Chart.js for the website	Builds the nine interactive charts and the final dashboard, and lets viewers see them online.
6.	Front-End / User Interface	Interface through which users interact with dashboards, filters, parameters, and visual insights.	Tableau Public dashboard; HTML website	Users interact with filters, tooltips and click-to-filter on Tableau Public or on the project website.